

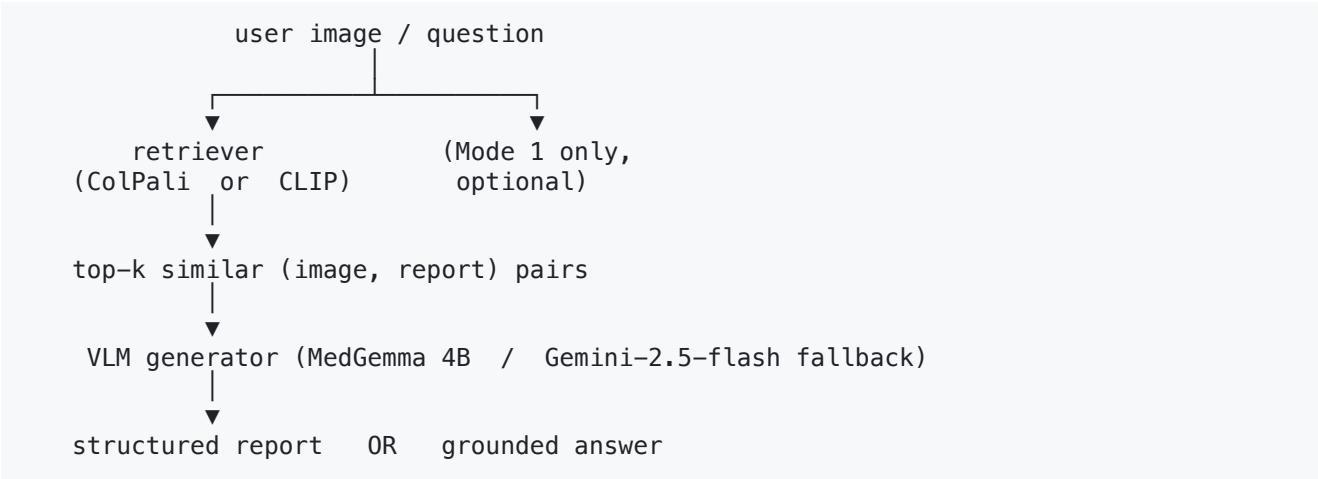
MedXray — Multi-Modal Chest X-Ray System

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1. System

Two independent modes sharing one model stack:

- **Mode 1 — Report Generation.** Image → structured radiology report (FINDINGS / IMPRESSION / RECOMMENDATIONS).
- **Mode 2 — RAG QA.** (Image + question) → grounded answer + retrieved evidence.



2. Models compared

Model	Role	Why it's here
ColPali v1.3 (PaliGemma + multi-vector head)	Retriever	Mandatory. Multi-vector late-interaction preserves region info — better fit for medical imagery than a single global vector.
CLIP ViT-B/32	Retriever (baseline)	Suggested. Fast single-vector reference for the comparison.
MedGemma-4B-IT	Generator	Mandatory. Medical-domain VLM. Used as the structured-report writer.
Gemini-2.5-flash	Generator fallback	Practical: lets the system run on a laptop without a GPU.

3. Dataset

- **Image-report corpus.** Per the assignment, MIMIC-CXR is the target. Since the Kaggle MIMIC dump is credentialed and didn't fit on my disk after the model downloads, I substituted a 40-image chest-imaging subset of the public-domain `unsloth/Radiology_mini` (ROCOv2). Loader in `scripts/load_radiology_mini.py`.
- **QA dataset (custom).** There is no QA set shipped with MIMIC-CXR, so I generated one. For each report I prompted **Gemini-2.5-flash** with the report only (image never shown) and

asked for 3 diverse Q/A/rationale triples that a clinician might realistically ask, mixing yes-no / location / severity / comparison / differential. Output is JSONL with `{image_id, image_path, question, answer, rationale, source_report}`. 68 pairs across 23 distinct images. Script: [src/data/create_qa_dataset.py](#).

4. Pipelines

- **Retrieval.** ColPaliStore (multi-vector, MaxSim score) and FlatCLIPStore (single-vector, cosine score). One Retriever abstraction exposes `search_by_image` and `search_by_text` so the QA mode can do a two-pass retrieval (image then question text).
- **Generation.** A single MedGemmaWrapper interface. A GeminiVLMWrapper mirrors it exactly so swapping is an env-var: `MEDXRAY_GENERATOR=medgemma | gemini | auto` (default `auto` tries MedGemma first, falls back to Gemini if it can't load).

5. Results

Retrieval evaluation (no API quota needed, fully reproducible)

40-image corpus, 60 QA questions over 20 distinct images, `k=5`.

Retriever	text→image R@1	text→image R@5	text MRR	image→image R@1	text query (s)	image query (s)
CLIP ViT-B/32	0.033	0.200	0.079	1.000	0.03	0.03
ColPali v1.3 (base only — see §6)	0.000	0.133	0.049	1.000	0.59	5.58

Image→image self-retrieval is perfect for both backends. That confirms the indices are sound.

Text→image is harder: CLIP at 20% beats the random-baseline of 12.5%; my ColPali run sits at random (see caveat §6). **Latency cost** of ColPali is ~20–180× CLIP per query.

End-to-end (single-image qualitative)

Smoke test on an echocardiogram showing a right atrial thrombus (`scripts/smoke_test.py`). With CLIP-RAG → Gemini:

Q: *Is there any evidence of cardiac abnormality in this image?* **A:** *Yes, there is evidence of a cardiac abnormality in this image. The echocardiogram shows a large echogenic mass (indicated by the white arrow) within the right atrium, which appears to extend into the right ventricle. This finding is consistent with a right atrial thrombus.*

Retrieval pulled the gold image at rank 1; end-to-end latency 12.15 s.

6. Honest limitations

1. **ColPali LoRA adapter was not loaded.** `colpali-engine 0.3.16` and `transformers 5.8` disagree on the language-model key prefix (`model.language_model.model.layers.*` vs `model.model.language_model.layers.*`). The LoRA weights are silently left at random init.

Practical fix: pin transformers < 5 . Numbers above are therefore "ColPali base", not the fine-tuned ColPali.

2. **No MIMIC-CXR.** Used the public ROCov2 subset instead (40 images).
3. **MedGemma was not run locally.** 4B at bf16 $\approx$ 8 GB weights + working RAM, and I had a 16 GB / 12 GB-free laptop. The wrapper is in place; MEDXRAY_GENERATOR=medgemma flips it on for a GPU/Colab run.
4. **Gemini free-tier quota** capped how many end-to-end generation evaluations I could run before submission. The retrieval table above is the result that doesn't depend on quota.
5. **Surrogate metrics.** BLEU/ROUGE measure surface form, not clinical correctness. A CheXbert-style label extractor would be a real next step.

7. Demo

Deploy



MedXray — Multi-Modal Chest X-Ray System

Two modes, three models compared. Choose a tab below. Built for the Multimedia Assignment 2 by **Ahmed Kahla** (ID: 202202231).

[Report Generation](#) [RAG QA](#) [About / Models](#)

Report Generation Mode

Upload a chest X-ray. The system writes a structured radiology report.

Chest X-ray

Upload

200MB per file • PNG, JPG

Optional clinical indication

e.g. cough, fever, post-op chest pain

Strategy ⓘ

☒ medgemma_direct ☐ colpali_rag ☐ clip_rag

Mode 1 — Report Generation. Image upload, retrieval strategy selector, generated structured report, parsed sections, and ranked retrieved references.



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Report Generation **RAG QA** About / Models

QA Mode (RAG)

Ask a clinical question about the uploaded X-ray. Answers are grounded in retrieved reports.

Chest X-ray

📁 Upload

200MB per file • PNG, JPG

Question

e.g. Is there evidence of pleural effusion?

Retriever

colpali

Top-k retrieved reports

4

Mode 2 — RAG QA. The model answers a clinical question using the image plus retrieved reports as grounded evidence.



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Report Generation RAG QA About / Models

Models used

Model	Role	Why
MedGemma 4B IT	image+text generation	medical-domain VLM — primary generator
ColPali v1.3	visual document retrieval	late-interaction multi-vector retriever
CLIP ViT-B/32	retrieval baseline	open-domain reference for comparison

Pipeline

- *Report mode*: image → (optional retrieval) → MedGemma → structured report.
- *QA mode*: (image + question) → retriever (image AND text passes) → MedGemma.

Disclaimer. Research/education only. Not a medical device.

About tab. The three model roles and the disclaimer.

8. Reproduce (laptop, no GPU)

```
python3.13 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
cp .env.example .env # add HUGGINGFACE_TOKEN + GOOGLE_API_KEY

python -m scripts.load_radiology_mini --limit 40 # data
python -m scripts.build_index --backend clip # ~5s
python -m scripts.build_index --backend colpali # ~10min
python -m src.data.create_qa_dataset --reports data/sample_reports.csv \
--out data/qa_dataset/qa.jsonl --per-report 3 --backend gemini
python -m scripts.eval_retrieval --k 5 # the $5 table
MEDXRAY_GENERATOR=gemini streamlit run app/streamlit_app.py
```

Repository: <https://github.com/a7mdka7la/medxray>